THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 5: **Measurement of Length and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The length of a table is 1.2 m. In centimetres it is: 1

(a) 1200 cm **(b)** 0.12 cm **(c)** 120 cm **(d)** 12 cm

2. Which motion is both periodic and linear (back and forth along a line)? 1

(a) a falling stone **(b)** a spinning top
(c) a mass vibrating on a spring **(d)** a wheel

3. The metal hook at the end of a measuring tape helps to: 1

(a) show the time
(b) make the tape bend
(c) make the tape heavier
(d) hold the tape against the edge of the object so that measuring starts from zero

4. 3500 m expressed in km is: 1

(a) 0.35 km **(b)** 3.5 km **(c)** 35 km **(d)** 350 km

5. A body moving along a circular track at constant speed shows: 1

(a) rest **(b)** linear motion **(c)** no motion **(d)** circular motion

6. Why is a ruler with clear markings better than one with worn markings? 1

(a) heavier **(b)** easier to read correctly
(c) costlier **(d)** looks better

7. A rope of 2.5 m is cut into 5 equal pieces. Each piece in cm is: 1

(a) 50 cm **(b)** 25 cm **(c)** 500 cm **(d)** 5 cm

8. The Earth rotating on its axis is: 1

(a) linear motion **(b)** no motion
(c) random motion **(d)** rotational motion

9. A student measures the length of a book as 24 cm. In metres it is: 1

(a) 24 m **(b)** 240 m **(c)** 2.4 m **(d)** 0.24 m

10. A thread 80 cm long is wound around a bottle exactly 4 times. The circumference of the bottle is: 1

(a) 20 cm **(b)** 320 cm **(c)** 4 cm **(d)** 80 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Convert 3.75 km into metres and centimetres. 2

12. Distinguish between rotation and revolution with an example. 2

13. Find the time period of a pendulum if it makes 30 oscillations in 60 s. 2

14. A ribbon is 4.5 m long. Pieces of 25 cm are cut. How many can be made? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ribbon 2.4 m long is cut into pieces of 15 cm. Find the number of pieces and the leftover length. 3

16. A pendulum takes 42 s for 21 oscillations. Find its time period and the number of oscillations in 1 minute. 3

17. A cyclist covers 15 km. Express in m and cm. 3

OR

A thread of 1.1 m wraps round a ball exactly 5 times. Find the circumference in cm. 3

18. Describe how a measuring tape is used to measure the waist of a person. 3

OR

Which units would you choose for: length of a pen, length of a river, thickness of a coin? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how you would measure the circumference of a cylindrical glass and its diameter, and verify the ratio. 5

OR

Explain with examples the concept of relative motion for
(a) a train passenger and
(b) the Sun as seen from Earth. 5

20. A student measures the length of a table five times: 120.2, 120.4, 120.1, 120.3, 120.5 cm. Find the mean length and explain why repetition helps. 5

OR

Describe an activity to show that a pendulum's time period depends on length. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND